

10. The Spark Gap

Air as a switch — and the lesson where the danger starts

Lesson 10 · about 60 minutes · **adult-operated from here on**

ADULT-OPERATED: MAINS-DERIVED HIGH VOLTAGE

From here on the apparatus contains voltages and currents that kill people every year. The adult owns the mains plug, the transformer and every decision about energizing. Nobody touches anything inside the enclosure until the supply is unplugged *and* the capacitor has been shorted with the grounding stick. Read the **Safety** sheet completely before this lesson, not during it.

What you need

- A piezo igniter from a gas lighter (safe, and the only source the boys handle)
- Two smooth machine-screw electrodes with adjustable spacing
- A dry wooden or PVC stand
- Feeler gauges or a ruler
- Safety glasses for everyone
- *Only for the adult:* the neon-sign transformer, in its enclosure

Predict first

Air is an excellent insulator. What has to be true for a spark to cross a gap at all — and if you double the gap, does it take twice the voltage, more than twice, or less?

Do it — the safe version first

1. Mount two rounded electrodes facing each other on a dry insulating stand, set to about 1 mm.
2. Connect a piezo igniter across them. Click it. A spark jumps, with a snap. A piezo makes thousands of volts at almost no current, so this is safe to handle and it demonstrates the whole principle.
3. Widen the gap in steps and keep clicking. At some spacing the piezo can no longer bridge it. Record that distance — you have just measured the breakdown voltage of air in units of piezo clicks.
4. **Sharp versus round.** Swap one rounded electrode for a needle. It now fires at a wider gap, and you may see a faint corona glow before any spark at all. Field lines crowd at a point, exactly as in Lesson 2.
5. **Listen and smell.** The snap is air heated so fast it explodes — that is sound made by a spark. The sharp smell afterwards is ozone. Ventilate.

What it means

Breakdown	Air is an insulator until the electric field rips electrons off its molecules. Those electrons accelerate, hit more molecules, free more electrons — an avalanche. In under a microsecond a column of air becomes a conductor.
The number	Dry air breaks down at roughly 30 kV per centimetre at sea level, so about 3 kV per millimetre. That is a rule of thumb, not a constant: humidity, temperature, altitude, electrode shape and surface roughness all move it.
Not quite linear	Doubling the gap needs a bit more than double the voltage, and below about a tenth of a millimetre the relationship reverses entirely — Paschen’s law. For our purposes: wider gap, higher firing voltage, and the gap spacing is therefore how you <i>set</i> the voltage the capacitor charges to before it fires.
Why it is a good switch	It closes in well under a microsecond and carries enormous current with almost no resistance. No mechanical or semiconductor switch you can build at home comes close. That speed is what makes the primary tank ring hard.
Quenching	Once the gap has fired it stays conducting until the current falls low enough for the hot channel to cool. Getting it to stop at the right moment — after the energy has crossed to the secondary but before it sloshes back, Lesson 9 — is the hardest part of tuning a coil. It is why serious builders use multiple gaps in series or blow air through them.

EVERYTHING CHANGES AT THIS LESSON

The piezo demonstration is safe. The neon-sign transformer is not, and it is the adult’s alone: 9–15 kV at 30–60 mA, against roughly 10 mA across the chest to stop a heart. The boys may watch a supervised firing from a stated distance and may do every part of the mechanical build with the machine unplugged and the capacitor shorted. They do not touch the plug, the switch, or anything inside the enclosure. Ozone, ultraviolet, impulsive noise and broadband radio interference all start here too. Everything on the **Safety** sheet applies from this moment.

The link to the coil

The gap sets the firing voltage, and $E = \frac{1}{2}CV^2$ from Lesson 7 turns that voltage into the energy of each shot. Widen the gap and each shot carries more energy but fires less often; narrow it and you get frequent weak shots. That one adjustment, plus the primary tap from Lesson 8, is most of what tuning a Tesla coil consists of.

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